# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA17**

---

**COURSE OUTCOME COVERED**

***CO4:*** *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

*Assessment Tool Weightage: 50%*

---

**Duration : 1 hour**

**QUESTIONS: 4** **Total Marks:40**

Annexure A

**ARTICLE REVIEW**

---

***Code of Conduct:***

*I Vasantha Nithi D Y (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

## ARTICLE REVIEW

---

**INSTRUCTIONS**

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

*Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.*

### Task 1: Article Summary

**(a)** Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

**(b)** Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

### Task 2: Methodology Analysis

**(a)** Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

**(b)** Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

### Task 3: Results & Critical Evaluation

**(a)** Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

**(b)** Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

**(c)** Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

### Task 4: Reflection & Learning Outcome

**(a)** State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

**(b)** If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

## RUBRICS FOR CALCULATION:

| **Criteria** | **Weightage (%)** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** |
|---|---|---|---|---|---|
| Understanding of Article | 20% | Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments | Shows good understanding with minor gaps | Partial understanding; misses key ideas | Lacks understanding of the article |
| Summary | 20% | Provides concise and accurate summary highlighting all key points | Covers most key points with minor omissions | Incomplete or partially accurate summary | Poor or incorrect summary |
| Critical Analysis | 25% | Provides insightful critique with strong reasoning and evaluation | Provides reasonable analysis with some justification | Superficial analysis; limited evaluation | No meaningful analysis |
| Use of Evidence | 15% | Effectively uses examples, data, or references to support review | Uses some supporting evidence with minor gaps | Limited or weak supporting evidence | No supporting evidence provided |
| Organization | 20% | Well-structured, clear, and coherent writing with proper language and flow | Generally clear with minor issues | Poor organization; lacks clarity | Disorganized and difficult to understand |

**CO4 – ASSESSMENT TOOL 3**

**Article Review**

**Selected Article**

**Authors:** Md Al-Imran and Shamim H. Ripon

**Title:** *Network Intrusion Detection: An Analytical Assessment Using Deep Learning and State-of-the-Art Machine Learning Models*

**Journal:** International Journal of Computational Intelligence Systems

**Year:** 2021

**Volume:** 14, Article 200

**DOI:** 10.1007/s44196-021-00047-4

The article was published on **5 December 2021** and is an open-access research article in Springer.

**DOI:** 10.1007/s44196-021-00047-4

---

**Task 1 – Article Summary**

**(a) Domain and ML Problem**

**Domain**

**Cybersecurity / Network Intrusion Detection**

**ML Problem**

The paper addresses the problem of automatically distinguishing **normal network traffic from attack traffic** using machine learning and deep learning models.

The authors compare traditional models such as **Decision Tree, SVM and KNN**, ensemble models such as **Random Forest and XGBoost**, and deep learning models including **FFNN, LSTM and GRU**.

---

**(b) Article Summary**

The paper investigates machine-learning-based network intrusion detection using the Kyoto University Honeypot System 2013 dataset. The authors aim to determine how traditional machine learning, ensemble learning, and deep learning models perform when identifying normal and malicious network traffic. The study applies a complete workflow involving data cleaning, categorical encoding, normalization, feature selection, class balancing, model training, and evaluation. Decision Tree, SVM, KNN, Random Forest, XGBoost, Feed Forward Neural Network, LSTM, and GRU models are evaluated. The research gap identified by the authors is that previous studies had limited use of newer boosting and recurrent deep-learning approaches on the Kyoto dataset. The authors therefore provide a broader comparative assessment combining traditional ML, ensemble methods, and deep learning. They also use LIME to provide explanations for Decision Tree predictions. The results show strong performance across the models, with Random Forest achieving the best accuracy among the reported models. The study demonstrates that machine learning can provide effective intrusion detection while also highlighting the importance of class balancing and explainability.

---

**Task 2 – Methodology Analysis**

**(a) ML Techniques and Dataset**

**Dataset**

The authors use the **Kyoto University Honeypot System 2013 dataset**.

- Dataset size: approximately **1.99 GB zipped**
- Data period: 2013
- Features: **24**
- Data types: numerical, quasi-continuous and categorical
- Data source: real honeypot network traffic
- Classes: Normal and Attack

The dataset is highly imbalanced, with approximately **98.64% attack traffic** and **1.36% normal traffic**.

**Preprocessing**

The authors performed:

1. Data cleaning
2. Duplicate removal
3. Label encoding
4. Standardization
5. Min-Max normalization
6. Discretization
7. Feature selection
8. Class balancing using **SMOTETomek**

After feature selection, **19 features** were used for the experiment.

**Algorithms**

The paper evaluates:

- Decision Tree
- SVM
- KNN
- Random Forest
- XGBoost
- Feed Forward Neural Network
- LSTM
- GRU

The Decision Tree uses **Gini Impurity** as its splitting criterion.

**(b) CO4 Mapping**

| CO4 Topic | Connection |
|---|---|
| **Inductive Learning** | Models learn intrusion patterns from labelled network traffic. |
| **Decision Trees** | Decision Tree is directly used for binary intrusion classification. |
| **Statistical/ML Learning** | SVM and KNN are used as traditional ML classifiers. |
| **Reinforcement Learning** | Not used in this article. |

**Methodological Strength**

A major strength is the **broad comparison of multiple ML and deep learning techniques** using the same cybersecurity problem. The authors also address class imbalance using SMOTETomek and apply LIME for explainability.

**Limitation**

The authors did **not use the complete dataset** because of limited computing resources. They explicitly identify this as a limitation and suggest future experiments using the full dataset and other datasets.

---

**Task 3 – Results & Critical Evaluation**

**(a) Key Results**

The reported results include:

| Model | Accuracy | Precision | Recall | F1 / MCC |
|---|---|---|---|---|
| SVM | 96.89% | — | — | MCC 93.89% |
| Decision Tree | **99.10%** | 99.70% | 98.47% | F1 99.14% |
| Random Forest | **99.17%** | — | — | MCC 98.33% |

| Model | Accuracy | Precision | Recall | F1 / MCC |
|---|---|---|---|---|
| KNN | 98.11% | 99.80% | ~96.36% | MCC 96.28% |
| XGBoost | — | — | — | F1 98.79% |
| FFNN | 96.99% | ~97% | ~97% | MCC 94.40% |
| LSTM | 96.78% | ~97% | ~97% | MCC 93.62% |
| GRU | 96.80% | ~97% | ~97% | MCC 93.65% |

The paper reports Random Forest as the strongest traditional ML model, with **99.17% accuracy**, while the Decision Tree achieved **99.10% accuracy** and **99.14% F1-score**.

The paper does **not report AUC-ROC as one of its main comparison metrics**, instead emphasizing accuracy, precision, recall, F1-score and MCC.

---

### (b) Critical Evaluation

The reported results are very strong, but they should be interpreted carefully.

**Potential Biases / Concerns**

1. **Class imbalance:** 98.64% of the data represents attack traffic, so accuracy alone can be misleading. The use of MCC and recall is therefore important.
2. **Partial dataset:** The authors used only a sample because of computing limitations, so performance may differ on the complete dataset.
3. **Dataset dependence:** Results obtained on Kyoto honeypot traffic may not directly generalize to modern enterprise networks.
4. **High accuracy:** Values around 99% are impressive, but additional testing on independent datasets would provide stronger evidence.

**Additional Experiments**

The research could be strengthened by:

- Testing on **UNSW-NB15 or CICIDS2017**
- Performing cross-dataset validation
- Reporting ROC-AUC and PR-AUC
- Testing against newer attack types
- Evaluating computational cost and inference latency

---

**(c) Real-World Applicability**

The techniques can be deployed in:

- Enterprise Intrusion Detection Systems
- Data-center security
- Cloud network monitoring
- IoT security
- Security Operations Centers (SOC)

**Industry Challenges**

**Data availability:** New attack types may not exist in historical datasets.

**Scalability:** Network traffic generates very large volumes of data.

**Computational cost:** Deep learning models require more computational resources.

**Class imbalance:** Rare attack types can be difficult to detect.

**Explainability:** Security analysts need to understand why an alert was generated.

The authors address explainability through **LIME**, which can provide explanations for individual Decision Tree predictions.

---

**Task 4 – Reflection & Learning Outcome**

**(a) Three Key Insights**

**1. Decision Trees are Interpretable**

The paper demonstrated that a Decision Tree can achieve very high classification performance while producing understandable decision rules.

**CO4 Connection:** Decision Tree learning.

**2. Class Imbalance Matters**

The dataset was highly imbalanced, and the authors used **SMOTETomek** to combine oversampling and undersampling.

**CO4 Connection:** Inductive learning and model evaluation.

**3. Accuracy Alone is Not Enough**

The paper uses **precision, recall, F1-score and MCC** alongside accuracy. This is important for imbalanced classification problems.

**CO4 Connection:** Classification model evaluation and statistical learning.

---

**(b) Proposed Extension**

I would extend the research by testing the trained models on a **cross-dataset setup**, such as training on the Kyoto dataset and testing on **UNSW-NB15 or CICIDS2017**.

**Justification**

This would test whether the models have actually learned general intrusion patterns rather than memorizing characteristics specific to one dataset.

**Expected Impact**

A model that maintains strong performance across different datasets would have greater potential for real-world deployment.

---

**Overall Conclusion**

The reviewed article demonstrates that machine learning and deep learning can be effectively applied to **network intrusion detection**. Decision Tree and ensemble models achieved particularly strong performance, while the use of class balancing and LIME improved the practical value of the system. However, the use of a sampled dataset and dependence on a single network dataset limit generalization. Testing across multiple modern datasets would make the findings more robust.

This covers **all four tasks and every sub-question** in Assessment Tool 3, including the article citation, 150–200 word summary, methodology, CO4 mapping, strengths/limitations, reported results, critical analysis, real-world applicability, three learning insights, and proposed extension.